

Student number:



EEE3094S 2025: Class Test 2

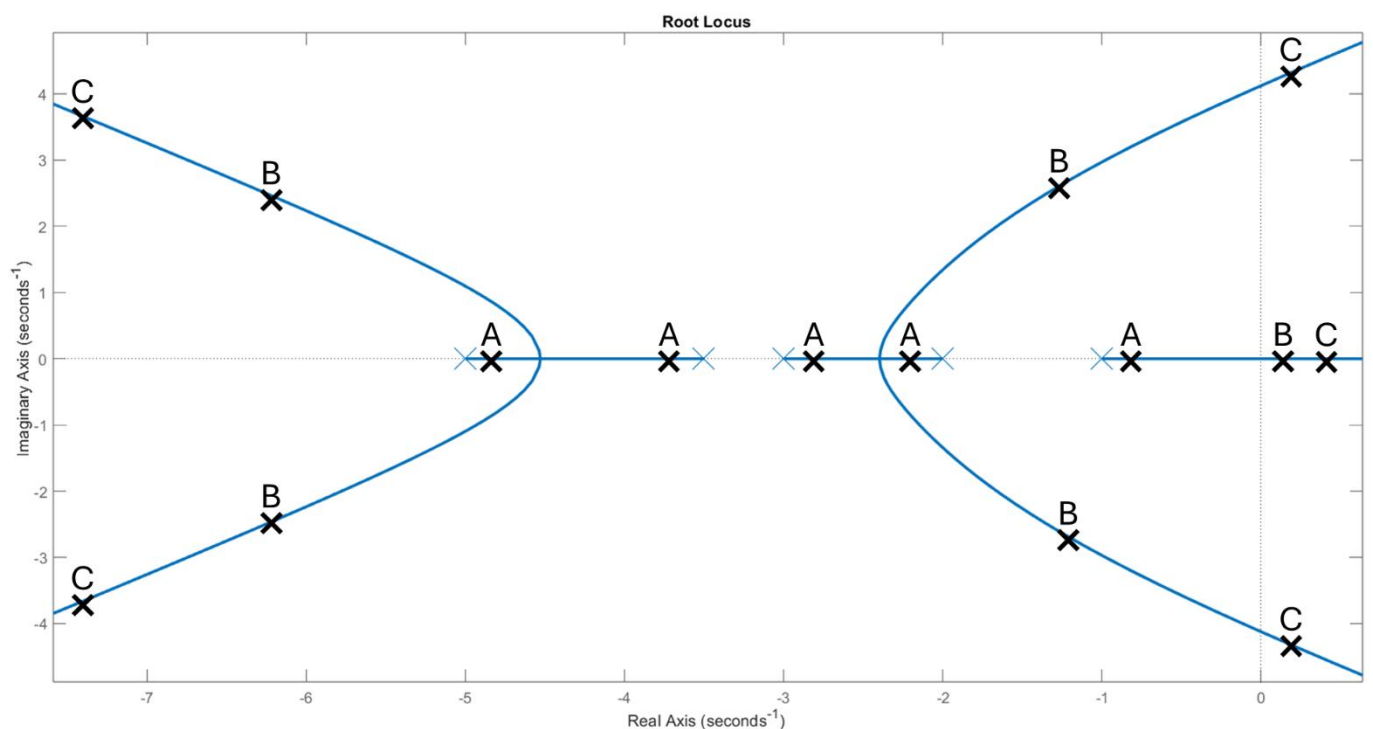
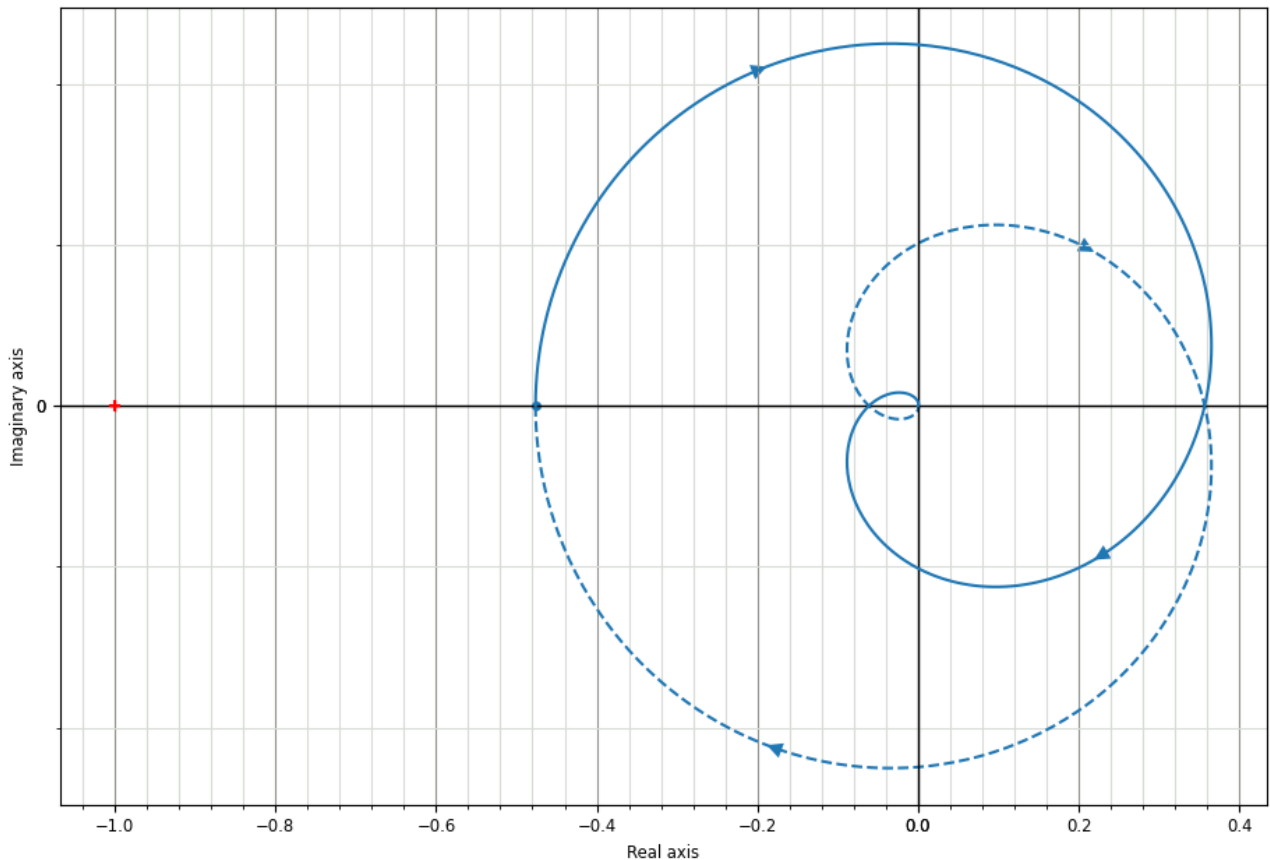
- The test is open book. You do not need to hand in your resources.
- Answer on the test sheet. If you run out of space, you may attach additional pages, but please note that you have done so (e.g. write “see attached pages” by the question) and make sure they are marked with your student number.
- You may do your diagrams in pencil but please ensure that your lines are clear.
- There are 4 questions worth a total of 50 marks.
- You have 90 minutes to complete the test + 5 minutes of reading time.

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Question 1 [9 marks]

The Nyquist plot and root locus of a system are shown below. The system is going to be placed in a negative feedback loop with a proportional controller. A few possible closed loop pole locations labelled A to C are marked on the root locus. For each one, use the Nyquist plot to determine an approximate range (“smaller than {value}”, “larger than {value} but smaller than {value}”, or “larger than {value}”) that the gain of the proportional controller would need to fall into for those to be the closed-loop poles. Explain your reasoning.

[3 x 3 marks]



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Question 1:

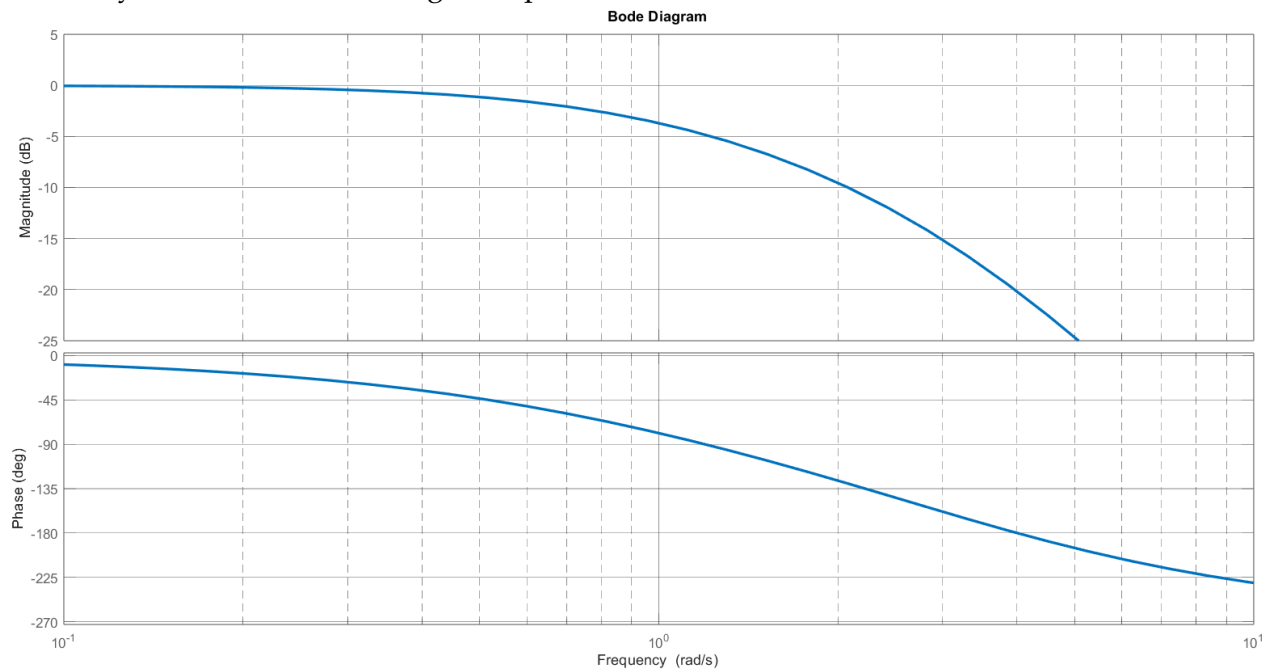
[illegible]

[illegible]

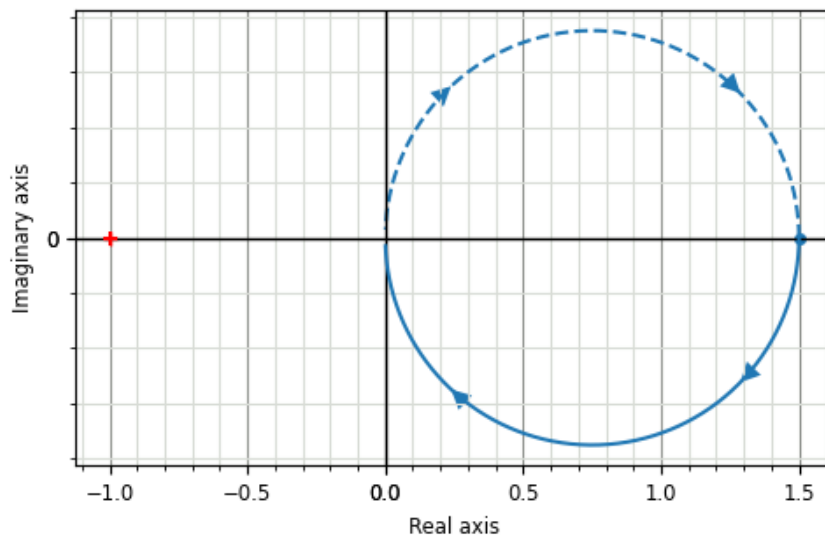
Question 2 [12 marks]

For each of the open-loop systems described, decide whether the system is stable in closed loop at its present value of gain, and if so, determine the gain margin of the system. Explain your answers.
[4 x 3 marks]

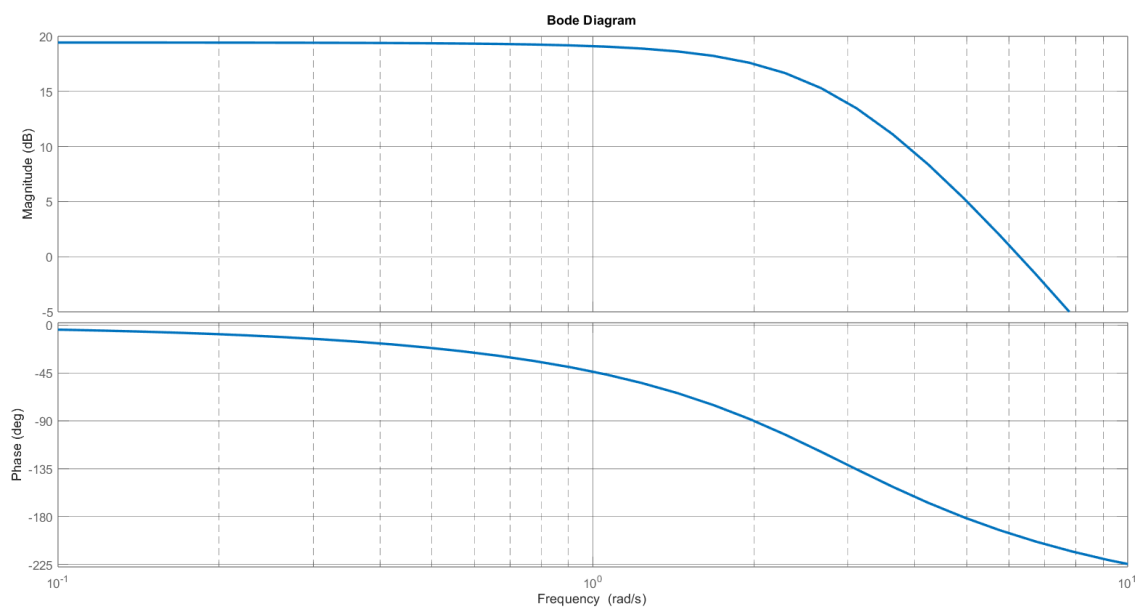
A) A stable system with the following Bode plot:



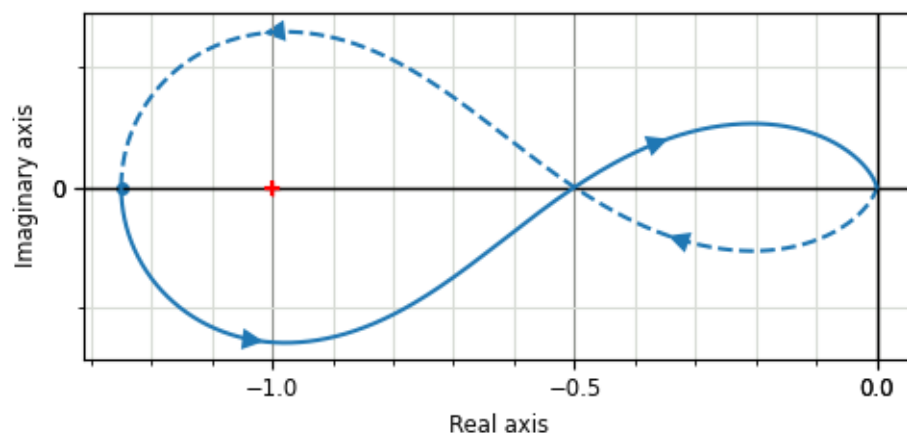
B) A stable system with the following Nyquist plot:



C) A stable system with the following Bode plot:



D) A system with one unstable open-loop pole and the following Nyquist plot:



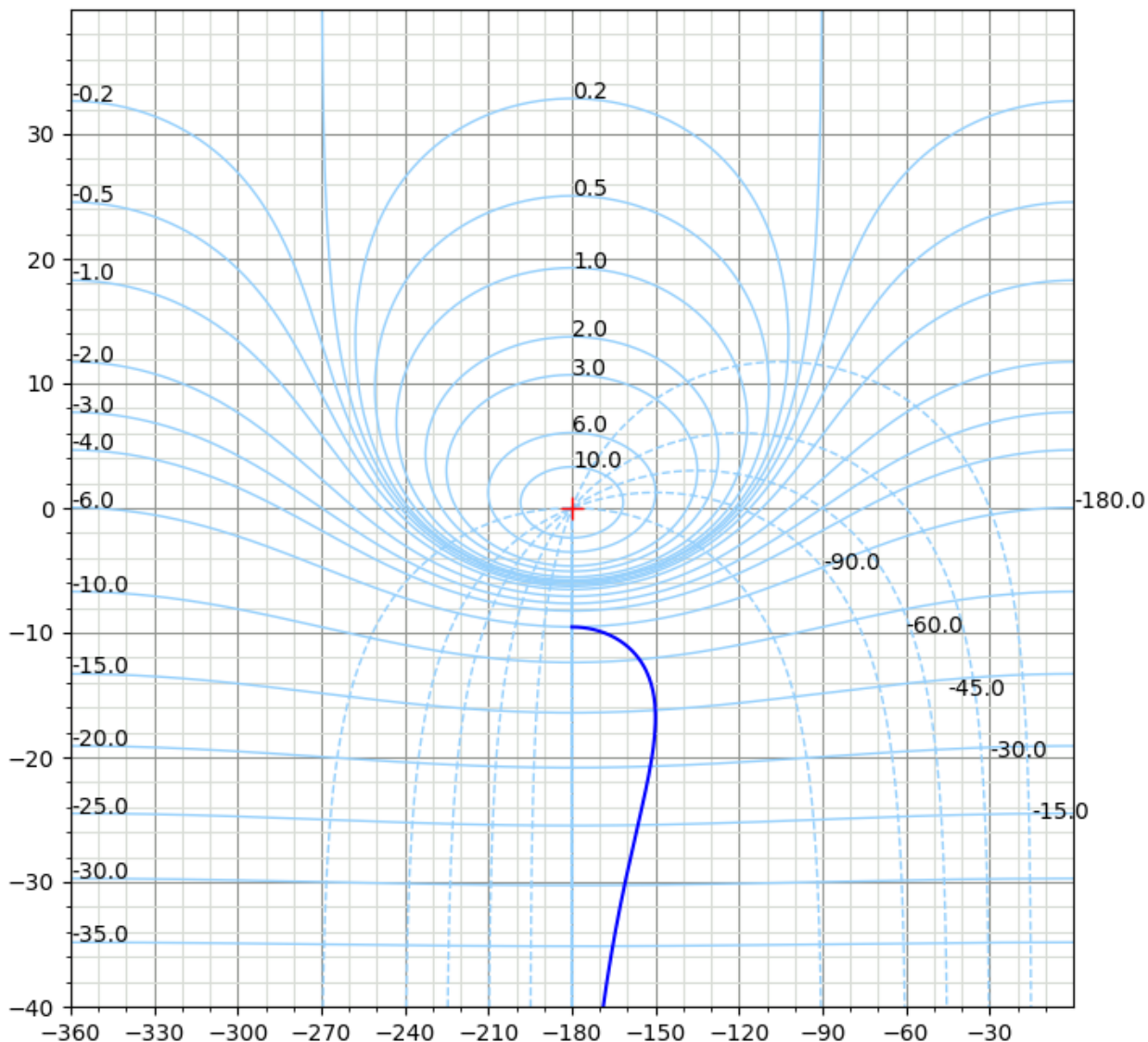
Question 2:

[illegible]

[illegible]

Question 3 [13 marks]

The Nichols chart of the unstable system $G(s) = \frac{1}{(s-1)(s+3)}$ is shown below.



- A) With reference to the Nichols chart, explain how you know it is possible to stabilize the system by placing it in negative feedback with a proportional controller, and determine the range of controller gains for which the closed-loop system will be stable. Indicate this on the Nichols chart by drawing a boundary. [4 marks]
- B) Draw boundaries on the plot representing the following:
 - i) A phase margin of 30 degrees [1 mark]
 - ii) Tracking accuracy of 90% with respect to constant setpoints. [2 marks]
- C) Is it possible to meet both the specifications in part B at once? You may draw on the plot to support your explanation. [3 marks]
- D) If you prioritize the accuracy specification in part B, ignoring all other specifications, give a best-case approximation for the percentage overshoot of the closed loop system. [3 marks]

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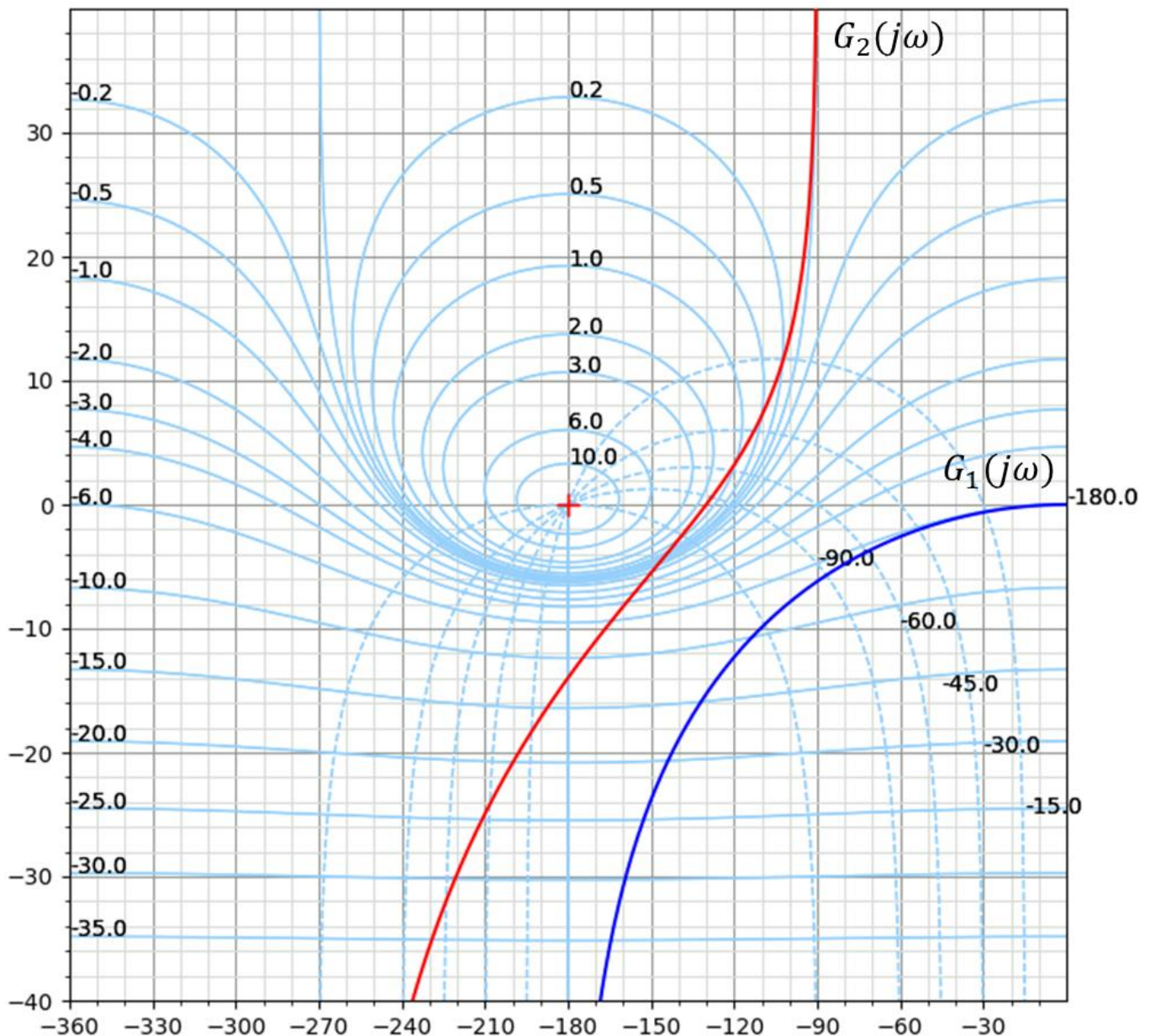
Question 3:

[illegible]

[illegible]

Question 4 [14 marks]

The frequency response of a system $G_1(s)$ is shown on the Nichols chart below. This system is to be placed in negative feedback with an integral controller of the form $\frac{K}{s}$. The open-loop frequency response of the system in cascade with an integrator $\frac{1}{s}$ is shown on the same plot as $G_2(j\omega)$.



- A) Does the addition of the integrator affect the gain margin of the system? [3 marks]
- B) The aim of placing the system in negative feedback is for it to satisfy the following specifications:
- Tracking position setpoints with accuracy of more than 95%.
 - Percentage overshoot of less than 20%.

Decide whether the system with the integral controller meets each of these specifications, supporting your answers with features from the Nichols chart. [2 x 3 marks]

- C) Your friend thinks it's unnecessary to go through the effort of implementing an integral controller, when it's possible to achieve the required tracking accuracy with a proportional controller.
- Determine the minimum gain required of this proportional controller, and re-draw the response

of the original system $G_1(j\omega)$ at this new gain. [2 marks]

(An additional chart is provided at the end of the answer space if you need one.)

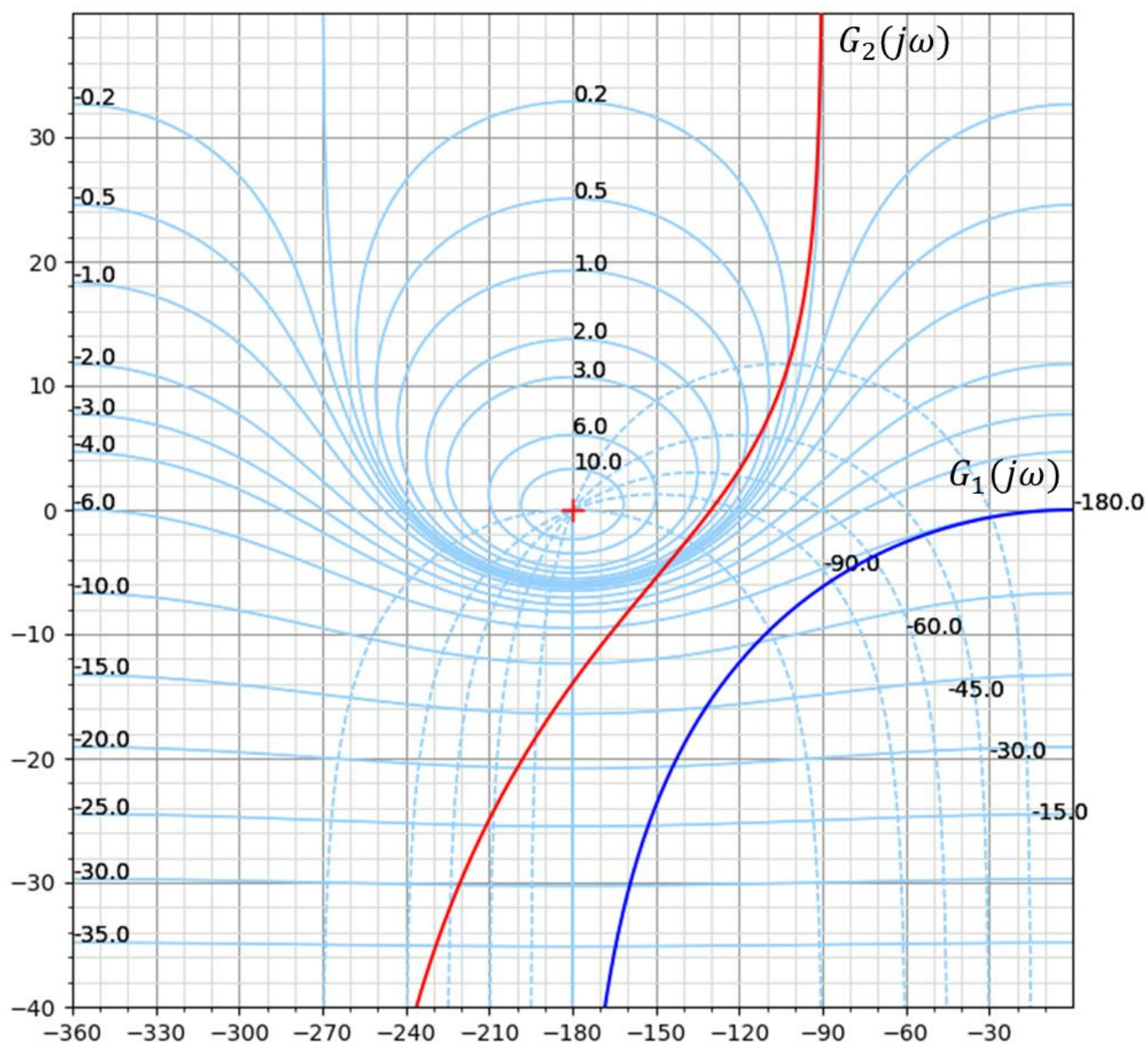
ii) Based on the response of the system with the proportional controller, would you agree with your friend that proportional control is the way to go? You may draw on/ annotate the Nichols chart further to support your answer. [3 marks]

Question 4:

[illegible]

[illegible]

Spare Nichols chart for q4.



Extra space:

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